



Original Article

The Role of Circulating Tumor Cell as a Promising Biomarker in the Evaluation of Pulmonary Nodules: A Prospective Study

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Purpose Our previous study showed that circulating tumor cell (CTC) count combined with gene mutation detection might help differentiate benign and malignant pulmonary nodules (PNs). Herein, we aimed to expand the study cohort and conduct further sequencing analysis.

Materials and Methods Patients with PNs were included, and CTCs were identified before operation. Low-coverage whole-genome sequencing (LC-WGS) and lung cancer-related targeted gene sequencing were performed on CTCs. The diagnostic efficacy was evaluated by receiver operating characteristic (ROC) curve. The differences in CTC counts among subgroups classified by demographic-clinical characteristics were analyzed. LC-WGS-based copy number variation (CNV) analysis and targeted gene mutation analysis were conducted.

Results A total of 172 patients were included. CTC count of 2.5 was identified by the ROC curves as the optimal diagnostic cutoff. The sensitivity and specificity of CTC count for differentiating benign and malignant PNs were 54.2% and 78.6%, respectively. The diagnostic sensitivity and specificity of combined CTC count, radiological nodule type, and any malignant imaging features were 84.7% and 71.4%, respectively. The CTC counts were significantly greater in patients with aggressive tumors, later stage, and spread through air spaces. CTCs from malignant cases had more CNVs than those from benign cases.

Conclusion CTC count can be used in identifying malignant PNs. The diagnostic efficacy can be improved if combined with computed tomography imaging characteristics. Further CNV analysis might help differential diagnosis. Greater CTC count might suggest more aggressive tumors. CTC detection can provide important information and guidance for subsequent management of PNs.

Key words Circulating tumor cell, Pulmonary nodule, Early diagnosis of lung cancer, Gene sequencing, Copy number variation

Introduction

Lung cancer is one of the most common and lethal malignant tumors worldwide [1]. The 5-year survival rate of lung cancer is closely related to the disease stage [2], which highlights the importance of early diagnosis. Low-dose spiral computed tomography (LDCT) is the commonly recommended method for early screening of lung cancer. With the extensive application of computed tomography (CT), the detection rate of pulmonary nodules (PNs) has increased significantly [3]. However, the accuracy of LDCT in distinguishing between benign and malignant PNs is limited,

which may lead to overdiagnosis and overtreatment [4]. Liquid biopsies for early cancer screening are highly regarded for their advantages, such as noninvasiveness and rapidity [5]. Circulating tumor cells (CTCs), which shed from tumor lesions into the peripheral blood, might be detected in the early stages of the disease [6]. However, the rarity of CTCs increases the difficulty of making the diagnosis. Because of the methodological and biological limitations, CTCs for early diagnosis of lung cancer have not been widely used in clinical practice [5].

Our previous study included patients with PNs and identified CTCs using modified isolation by size of epithelial

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tumor cell technique (ISET) [7]. We found significant differences in CTC counts between malignant and benign cases. However, CTCs were found in 80% of benign cases. Physical separation methods based on cell morphology do not rely on specific surface markers and can capture a broader range of CTCs. It's difficult to distinguish tumor cells from morphologically similar normal cells. To compensate for this flaw, single-cell sequencing technology might be used to determine the origin of CTCs and obtain rich biological genetic information [8]. Therefore, we previously performed whole exome sequencing in CTCs from six malignant cases and two benign cases. For instance, three specific single nucleotide variations (SNVs) of *TP53* were identified exclusively in CTCs from all four malignant cases, except for two malignant samples with low sequencing coverage [7]. Our previous results supported the possibility for CTCs combined with sequencing analysis to distinguish between benign and malignant PNs.

To further clarify whether the CTCs found in malignant and benign cases are different, we intend to perform more sequencing analysis on CTCs. However, the challenge lies in the fact that current whole-genome amplification methods for CTCs do not achieve sufficient accuracy by whole exome sequencing [9]. Alternatively, multiple annealing and looping-based amplification cycles (MALBAC) combined with low-coverage whole-genome sequencing (LC-WGS) for copy number variation (CNV) analysis of CTCs has been demonstrated in the context of coverage breadth, uniformity, and reproducibility [9].

Based on our previous study, we expanded the study cohort, and selected more efficient and cost-effective sequencing method, such as LC-WGS-based CNV analysis and lung cancer-related targeted gene sequencing. Moreover, the associations of CTC count with clinical factors and prognosis were investigated. In this study, we aimed to further clarify how to use CTC combined with gene sequencing to improve the efficacy for distinguishing between benign and malignant PNs and explore more possible roles of CTC detection in the early diagnosis of lung cancer.

Materials and Methods

1. Study design and participants

This study was a clinical trial conducted at China-Japan Friendship Hospital, and it was registered in ClinicalTrials.gov (NCT06187935). A total of 122 patients were included in the previous study from January 2019 to November 2022 [7]. Then, 50 patients were further included from April 2023 to November 2023. All patients had clinically suspected malignant PNs (≤ 30 mm) on chest CT, and received surgery or

biopsy. The full inclusion and exclusion criterion are shown in Supplementary Material.

According to the density shown on CT image, the PNs were classified into ground glass nodule (GGN), partial solid nodule (PSN), and solid nodule (SN) (definitions shown in Supplementary Material). All patients were diagnosed by histopathological examination and graded in line with the Ninth edition of tumor, lymph node, metastasis (TNM) classification [2]. Adenocarcinoma *in situ* (AIS), minimally invasive adenocarcinoma (MIA), invasive adenocarcinoma (IAC), and other malignant lung tumors (squamous cell carcinoma, small cell lung cancer, signet-ring cell carcinoma, and adenoid cystic carcinoma) were included in the malignant group. According to the World Health Organization classification of lung tumors, spread through air spaces (STAS) was defined as the spread of micropapillary clusters, solid nests, or single cells into the airspaces beyond the edge of the main tumor [10]. Blood samples of the enrolled patients were collected before operation. Follow-ups were carried out for the first batch of patients included in the study, with the last follow-up date being March 31, 2024. Recurrence was defined as local recurrence, regional recurrence, or else distant metastatic recurrence detected by imaging during follow-up after operation.

2. CTC detection and laser capture microdissection

CTCs were isolated from 5 mL peripheral blood by a modified ISET platform (CTCBIOPSY-A10, Wuhan YZY Medical Science and Technology Co., Ltd.), and identified according to morphological characteristics. Briefly, CTCs were diagnosed if the isolated cells met no less than four diagnostic criteria, or had large nucleoli or abnormal nuclear divisions along with two of the other criteria. The criteria are as follows: abnormal karyotype; nuclear-cytoplasmic ratio > 0.8 ; cell diameter > 15 μm ; deeply stained and unevenly colored nuclei; thickened nuclear membrane with depressions or folds, resulting in an irregular nuclear membrane; and large nucleoli. Fifteen cases were selected, and the LMD7000 system (Leica) was used for single-cell laser capture microdissection of the filter membrane. An ultraviolet laser was employed for the automated scanning and precise cutting of the identified CTC margins. Then, these specific cells were isolated and deposited into a collection tube. The same procedure was used to capture background white blood cells (WBCs) of each case.

3. Genome sequencing

DNA of captured cells was extracted and amplified using MALBAC method (MALBAC Single Cell WGA Kit, Yikon Genomics). The genomic DNA was randomly sheared into fragments around 350 bp for LC-WGS and 150-250 bp for tar-

geted gene sequencing, using the Covaris sonicator (Covaris Inc.). End repair and A-tailing were performed on the DNA fragments, and then adapters were ligated to both ends of the fragments to prepare the DNA library. The quality was assessed using Qubit 2.0 (Thermo Fisher Scientific Inc.) and Agilent 2100 (Agilent Technologies Inc.). The concentration of the library was accurately quantified by quantitative polymerase chain reaction (effective concentration > 3 nM). Besides, the target region was captured by a designed kit (TargetSeq One Hyb & Wash Kit with Eco Universal Blocking Oligo [for Illumina], Target probes IGTT506V3). The capture probes were designed for lung-cancer-related targeted genes (*ALK*, *BCL2L11*, *BRAF*, *EGFR*, *ERBB2*, *KRAS*, *MAP2K1*, *MET*, *NRAS*, *NTRK1*, *NTRK2*, *NTRK3*, *PIK3CA*, *RET*, *ROS1*, and *TP53*). The sequencing was performed on the qualified sequence using an Illumina NovaSeq 6000 system (Illumina).

4. Bioinformatics analysis

First, the raw sequencing data were processed for quality control and filtering using the fastp and SOAPnuke to obtain clean data. Then, the clean data were mapped to human genome 19 (hg19, GRCh37) using bwa. Duplicate marking, local realignment, and base quality recalibration were conducted by Picard, SAMtools, and GATK to generate high-quality BAM files. Using the LC-WGS data, the average genomic coverage for each 200-kb bin was calculated, and the coverage for each bin of CTCs was normalized as Z-score with the WBCs as the control [11]. Next, the R package “DNAcopy” based on circular binary segmentation was used to identify the significant breakpoints and obtain the significantly changed segments. The Z-scores of each segment were calculated accordingly. Besides, somatic SNVs and insertions and deletions (InDels) were detected by the Mutect2 module in GATK using the targeted gene sequencing data. Mutations were annotated by ANNOVAR. And mutations shared between CTCs and patient-matched WBCs were filtered out before subsequent mutation analyses on CTCs. The relationship between the detected variations and diseases was queried in the ClinVar database (<https://www.ncbi.nlm.nih.gov/clinvar/>).

5. Statistical analysis

The chi-square test or the Fisher’s exact test was used to compare the distribution of demographic-clinical characteristics between the benign group and the malignant group. The Mann-Whitney U test or the Kruskal-Wallis test was used to compare CTC counts among different subgroups, and Bonferroni correction was used for pairwise comparisons among multiple subgroups. Categorical variables were described with absolute frequency counts and percentages, while continuous variables were described with medians and

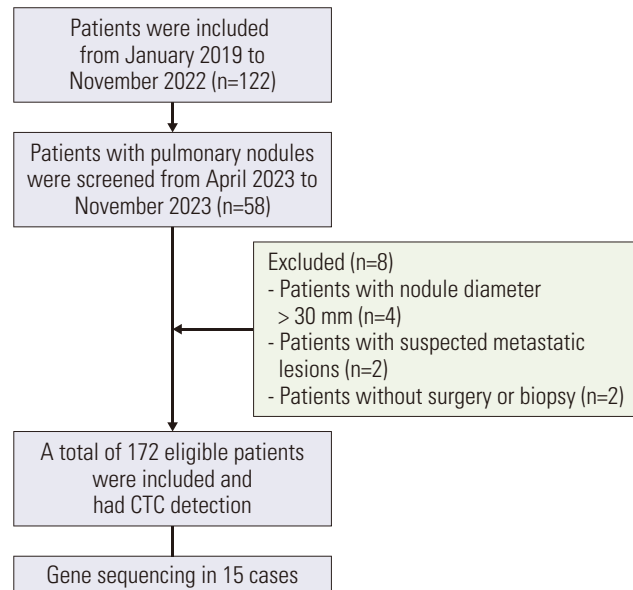


Fig. 1. Flowchart of including patients. CTC, circulating tumor cell.

interquartile ranges (IQRs). Receiver operating characteristic (ROC) curve analysis was used to determine the optimal cut-off of CTC count for the classification of benign and malignant PNs. The corresponding sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were subsequently calculated. Binary logistic regression was used to identify the risk factors associated with the diagnosis of malignant PNs. Age group, sex, smoking history, chronic lung disease history, lung cancer family history, radiological nodule type, nodule diameter, any malignant imaging features, tumor biomarker, and CTC count grouped by 2.5 cells were separately included in the logistic regression model. Then, the variables that showed statistical significance in the univariate analysis were included simultaneously in the logistic regression model, and the forward stepwise logistic regression approach was used. The variables ultimately retained in the model were significantly associated with the diagnosis. The value of the combination of the significantly associated variables for diagnosis was further explored utilizing ROC curve analysis. Patients with any cause of death or those who were alive were censored at the time of death or the last follow-up. Cumulative survival rate was calculated by Kaplan-Meier analysis. p -value < 0.05 (two-sided probability) was considered statistically significant. Analyses were conducted and the figures were generated using SPSS ver. 26 (IBM Corp.), GraphPad Prism ver. 9.5 (GraphPad Software), Origin 2024 (OriginLab Corporation), and R ver. 4.4.0 (R Foundation for Statistical Computing).

Table 1. Baseline characteristics of the included patients

Variable	Total (n=172)	Benign group ^{a)} (n=28)	Malignant group ^{a)} (n=144)	p-value ^{b)}
Age (yr)				
< 50	38 (22.1)	11 (39.3)	27 (18.8) ^{c)}	0.017
≥ 50	134 (77.9)	17 (60.7)	117 (81.3)	
Sex				
Male	68 (39.5)	13 (46.4)	55 (38.2)	0.415
Female	104 (60.5)	15 (53.6)	89 (61.8)	
Smoking history				
Yes	43 (25.0)	9 (32.1)	34 (23.6)	0.340
No	129 (75.0)	19 (67.9)	110 (76.4)	
Chronic lung disease				
Yes	24 (14.0)	6 (21.4)	18 (12.5)	0.342
No	148 (86.0)	22 (78.6)	126 (87.5)	
Lung cancer family history				
Yes	19 (11.0)	4 (14.3)	15 (10.4)	0.789
No	153 (89.0)	24 (85.7)	129 (89.6)	
Radiological nodule type				
GGN	50 (29.1)	3 (10.7) ^{c)}	47 (32.6)	< 0.001
PSN	86 (50.0)	9 (32.1)	77 (53.5)	
SN	36 (20.9)	16 (57.1)	20 (13.9)	
Nodule diameter (mm)				
< 6	2 (1.2)	1 (3.6) ^{c)}	1 (0.7)	0.261
≥ 6, < 8	20 (11.6)	5 (17.9)	15 (10.4)	
≥ 8, < 15	71 (41.3)	11 (39.3)	60 (41.7)	
≥ 15	79 (45.9)	11 (39.3)	68 (47.2)	
Any malignant imaging features				
Yes	132 (76.7)	14 (50.0)	118 (81.9)	< 0.001
No	40 (23.3)	14 (50.0)	26 (18.1)	
Tumor biomarker				
Normal	95 (55.2) ^{d)}	16 (57.1) ^{d)}	79 (54.9) ^{d)}	0.266
Elevated	66 (38.4)	7 (25.0)	59 (41.0)	
Histological subtype				
AIS	12 (7.0) ^{c)}	N/A	12 (8.3)	N/A
MIA	43 (25.0)	N/A	43 (29.9)	
IAC	83 (48.3)	N/A	83 (57.6)	
Other malignant tumors ^{a)}	6 (3.5)	N/A	6 (4.2)	
AAH	1 (0.6)	1 (3.6)	N/A	
Hamartoma	3 (1.7)	3 (10.7)	N/A	
Pulmonary inflammation ^{a)}	13 (7.6)	13 (46.4)	N/A	
Pulmonary fibroplasia/Fibrosis ^{a)}	9 (5.2)	9 (32.1)	N/A	
Carbon deposition nodule	1 (0.6)	1 (3.6)	N/A	
Pneumoconiosis	1 (0.6)	1 (3.6)	N/A	
STAS ^{e)}				
Yes	13 (7.6) ^{d)}	N/A	13 (9.0) ^{d)}	N/A
No	74 (43.0)	N/A	74 (51.4)	
LVI ^{e)}				
Yes	11 (6.4) ^{d)}	N/A	11 (7.6) ^{d)}	
No	76 (44.2)	N/A	76 (52.8)	

(Continued to the next page)

Table 1. Continued

Variable	Total (n=172)	Benign group ^{a)} (n=28)	Malignant group ^{a)} (n=144)	p-value ^{b)}
Stage				
Tis	12 (7.0) ^{d)}	N/A	12 (8.3) ^{c)}	N/A
Ia1	56 (32.6)	N/A	56 (38.9)	
Ia2	48 (27.9)	N/A	48 (33.3)	
Ia3	11 (6.4)	N/A	11 (7.6)	
Ib	10 (5.8)	N/A	10 (6.9)	
≥ IIa	7 (4.1)	N/A	7 (4.9)	
CTC count				
0	25 (14.5) ^{c)}	6 (21.4) ^{c)}	19 (13.2)	0.006
1-2	63 (36.6)	16 (57.1)	47 (32.6)	
> 2	84 (48.8)	6 (21.4)	78 (54.2)	

AAH, atypical adenomatous hyperplasia; AIS, adenocarcinoma *in situ*; CTC, circulating tumor cell; GGN, ground glass nodule; IAC, invasive adenocarcinoma; LVI, lymphovascular invasion; MIA, minimally invasive adenocarcinoma; N/A, not applicable; PSN, partial solid nodule; SN, solid nodule; STAS, spread through air spaces. ^{a)}Benign group: AAH, hamartoma, pulmonary inflammation, pulmonary fibroplasia/fibrosis, carbon deposition nodule, pneumoconiosis; malignant group: AIS, MIA, IAC, other malignant tumors; other malignant tumors: squamous cell carcinoma, small cell lung cancer, adenoid cystic carcinoma, signet-ring cell carcinoma; pulmonary inflammation: granulomatous inflammation, organizing pneumonia, infectious inflammation, etc.; pulmonary fibroplasia/fibrosis: seven cases were diagnosed as pulmonary fibroplasia accompanied by chronic inflammation, one case was diagnosed as pulmonary fibroplasia, and one case was diagnosed as pulmonary focal fibrosis. ^{b)}p-values were derived from the chi-square test or the Fisher's exact test. ^{c)}The data were rounded to one decimal place. The sum of the proportions of each subgroup did not equal 100%, and this was a normal error resulting from rounding. ^{d)}Due to missing or inapplicable data for some cases, the total proportion did not amount to 100%. ^{e)}STAS and LVI were studied in patients with IAC or other malignant tumors. Two patients did not have the information about STAS and LVI, as they received puncture biopsy only.

Results

1. Baseline characteristics and follow-up

A total of 172 patients were included in the study (Fig. 1), with a median age of 59.0 years old (data not shown). Twenty-eight cases had benign diseases, including atypical adenomatous hyperplasia, hamartoma, pulmonary inflammation, pulmonary fibroplasia/fibrosis, carbon deposition nodule, and pneumoconiosis. Twelve cases had AIS, 43 cases had MIA, 83 cases had IAC, and six cases had other lung malignant tumors. Most patients were female (60.5%), nonsmokers (75.0%), without family history of lung cancer (89.0%) or chronic lung disease history (86.0%, including chronic bronchitis, emphysema, interstitial lung abnormalities/diseases, and tuberculosis). The median follow-up duration was 32.0 months (95% confidence interval [CI], 29.1 to 34.9), the 5-year overall survival rate was 96.8%, and the 5-year disease-free survival rate was 84.7% (S1 Fig.).

The distributions of baseline characteristics classified by benign and malignant diseases are presented in Table 1. The proportion of malignant diseases in patients older than 50 years was significantly higher than that in patients younger than 50 years (87.3% vs. 71.1%, $p=0.017$). There were no significant differences in other demographic characteristics

between the benign and the malignant groups. PSN was the most common (50.0%) type, with a high proportion of malignant diseases (89.5%). The smallest number of SNs (20.9%) exhibited the highest proportion of benign diseases (44.4%). The proportion of malignant diseases increased with larger nodule size. Most patients (76.7%) had at least one malignant imaging sign (including lobulation, spiculation, pleural indentation, vacuole sign, cavity, air bronchogram sign, vascular convergence sign, halo sign), these cases had more malignant diseases than those without (89.4% vs. 65.0%, $p < 0.001$). Preoperative lung cancer tumor biomarkers (neuron specific enolase, carbohydrate antigen [CA] 72-4, CYFRA21-1, α -fetoprotein, carcinoembryonic antigen, CA-125, CA19-9, CA15-3, progastrin-releasing peptide, and squamous cell carcinoma antigen) were available in 161 patients, and they were not elevated in 95 patients. There was no statistically significant difference between the benign and malignant groups regarding the elevation of tumor biomarkers. The comparisons in each tumor biomarker between these two groups are presented in S2 Table.

2. Diagnostic value of CTC count for PNs

Consistent with our previous study, CTC count of 2.5 per 5 mL of peripheral blood was identified by the ROC curve

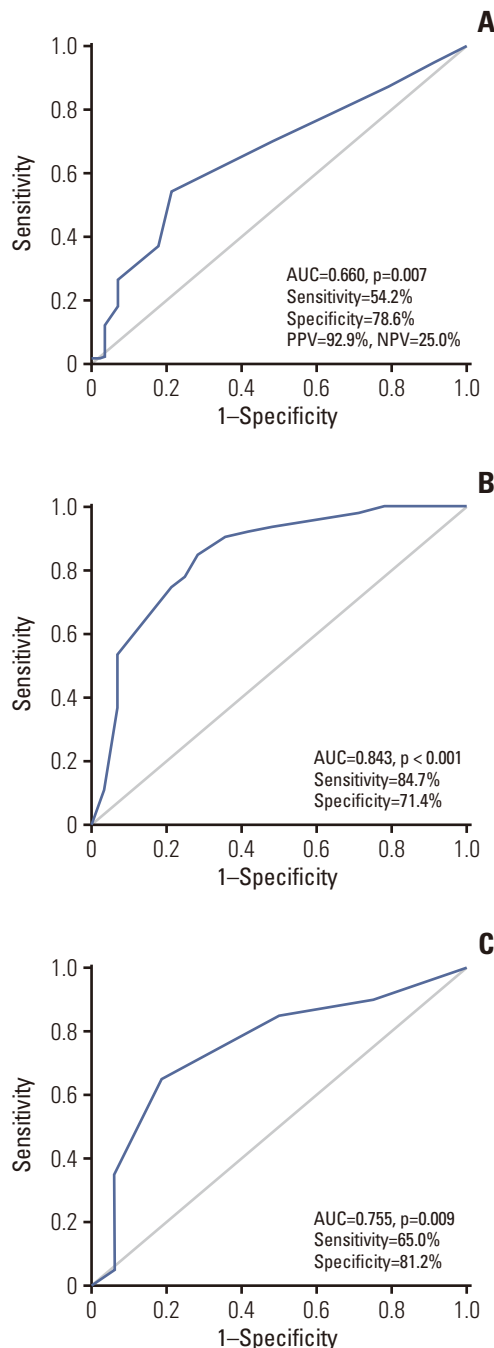


Fig. 2. The diagnostic efficacy of circulating tumor cell (CTC) count for patients with pulmonary nodules (PNs). (A) Receiver operating characteristic (ROC) curve for distinguishing between benign and malignant PNs by CTC count. (B) Multifactor combined diagnostic ROC curve for distinguishing between benign and malignant PNs. (C) ROC curve for distinguishing between benign and malignant PNs by CTC count in the solid nodule subgroup. The factors for combined diagnosis were radiological nodule type, any malignant imaging features, and CTC count grouped by 2.5 cells. AUC, area under the curve; NPV, negative predictive value; PPV, positive predictive value.

as the optimal cutoff value for the following diagnostic tests. The area under the curve (AUC) of CTC count for the differential diagnosis of benign and malignant PNs was 0.660 ($p=0.007$). The sensitivity and specificity were 54.2% and 78.6%, and the PPV and NPV were 92.9% and 25.0%, respectively (Fig. 2A). Based on the cutoff value of CTC count, the patients were classified into three subgroups (0, 1-2, and > 2). The differences in the distribution of benign and malignant cases among the three subgroups were statistically significant ($p=0.006$) (Table 1).

The results of univariate logistic regression analysis are shown in S3 Table. The factors retained in the multivariate logistic regression model for combined diagnosis were radiological nodule type, any malignant imaging features, and CTC count grouped by 2.5 cells. CTC count greater than 2.5 (odds ratio [OR], 6.0; 95% CI, 1.9 to 18.9; $p=0.002$), non-SN (GGN: OR, 24.4; 95% CI, 5.2 to 115.2; $p < 0.001$; PSN: OR, 7.0; 95% CI, 2.4 to 20.5; $p < 0.001$), and the presence of any malignant imaging features (OR, 6.5; 95% CI, 2.2 to 19.0; $p=0.001$) were the independent risk factors for malignant PNs (S3 Table). The AUC of the multifactor combined diagnostic ROC curve for distinguishing between benign and malignant PNs was 0.843, and the sensitivity and specificity were 84.7% and 71.4%, respectively (Fig. 2B).

Further classifying by radiological nodule type, although CTC count greater than 2.5 suggested an increased risk of malignant PNs in all subgroups, this was statistically significant only in the SN subgroup (OR, 8.0; 95% CI, 1.7 to 38.1; $p=0.009$) (S4 Table). CTC count showed the best diagnostic efficacy in the SN group (AUC, 0.755; $p=0.009$; sensitivity, 65.0%; specificity, 81.2%) (Fig. 2C). The AUC of CTC count in the GGN subgroup and the PSN subgroup was 0.543 and 0.654, respectively ($p > 0.05$, data not shown).

3. Relationship between CTC count and demographic-clinical characteristics

Fig. 3 shows the comparison of CTC counts among different subgroups. There was no significant difference in CTC counts between the subgroups classified by age and sex (Fig. 3A and B). The CTC count elevated as the nodule diameter increased, but the differences were not statistically significant ($p=0.314$) (Fig. 3C). The CTC counts in the IAC/other malignant tumor group were significantly greater than those in the benign disease group (median [IQR], 3.0 [1.0 to 5.0] vs. 1.5 [1.0 to 2.0]; $p=0.011$) (Fig. 3D). The CTC counts in both MIA and AIS group were also slightly greater than those in the benign disease group (median [IQR], 2.0 [1.0 to 4.0] vs. 2.0 [2.0 to 3.0] vs. 1.5 [1.0 to 2.0]). However, no significant differences were observed ($p > 0.05$). In the subgroup of IAC and other malignant tumors, the CTC counts in patients with STAS were significantly greater than those in patients with-

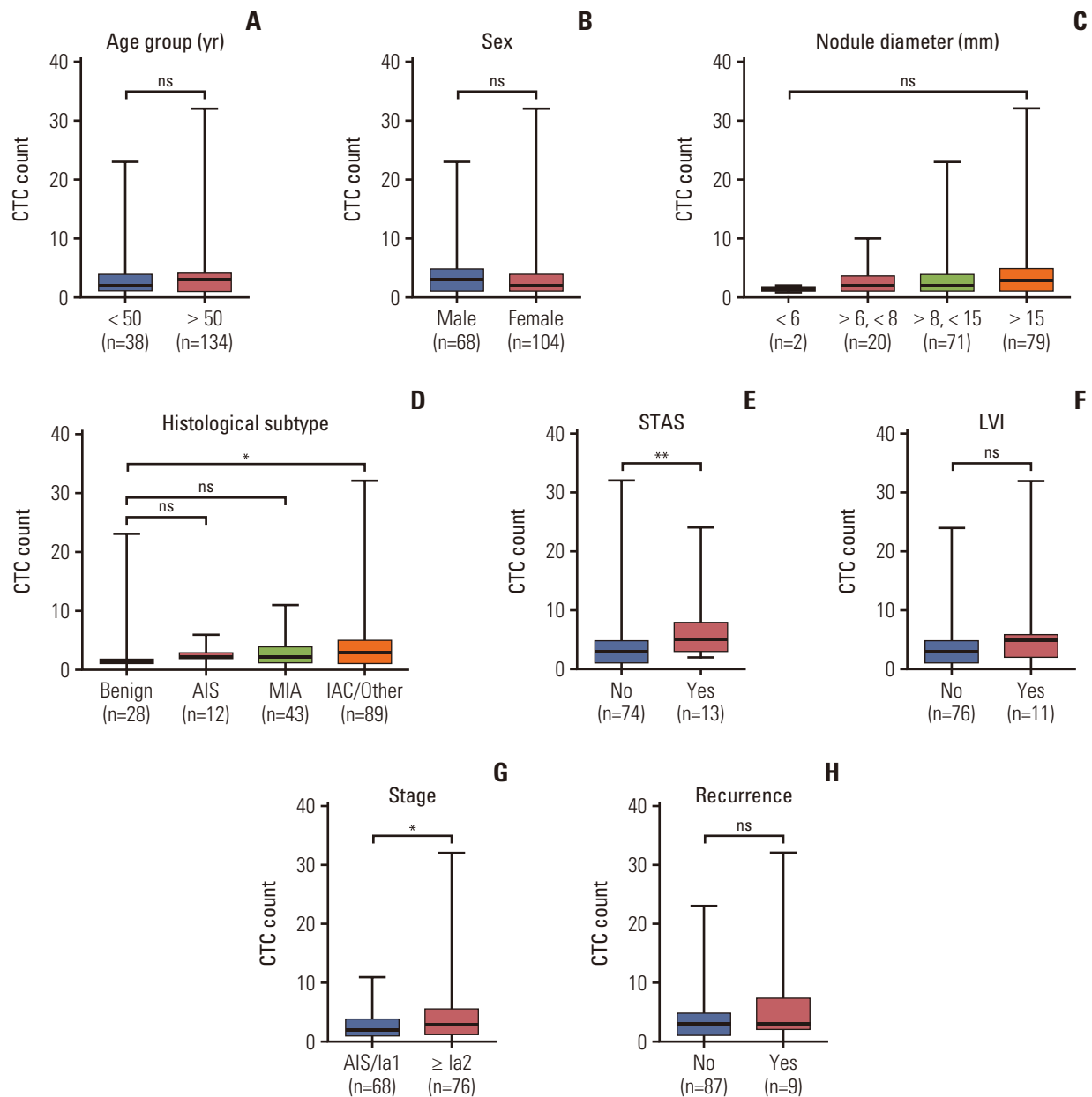


Fig. 3. The comparison of circulating tumor cell (CTC) counts among different subgroups. Differences in CTC counts among patients in different age groups (A), with different sex (B), with nodules of different diameters (C), with different histological subtypes (D), with or without spread through air spaces (STAS) (E) and lymphovascular invasion (LVI) (F), with different stages (G), and with or without recurrence (H). ns, $p > 0.05$; * $p < 0.05$; ** $p < 0.01$. p-values were derived from the Mann-Whitney U test or the Kruskal-Wallis test or the Bonferroni correction method. The Box-and-Whiskers plot was used for data visualization. AIS, adenocarcinoma *in situ*; IAC/other, invasive adenocarcinoma and other lung malignant tumors; MIA, minimally invasive adenocarcinoma.

out (median [IQR], 5.0 [3.0 to 8.0] vs. 3.0 [1.0 to 5.0]; $p=0.008$) (Fig. 3E). Patients with lymphovascular invasion (LVI) also had greater CTC counts, but the difference was not statistically significant (median [IQR], 5.0 [2.0 to 6.0] vs. 3.0 [1.0 to 5.0]; $p=0.202$) (Fig. 3F). The patients with later stage disease ($\geq$ stage Ia2, $n=76$) had significantly greater CTC counts

than those with earlier stage disease (Tis and stage Ia1, $n=68$) (median [IQR], 3.0 [1.3 to 5.8] vs. 2.0 [1.0 to 4.0]; $p=0.026$) (Fig. 3G). In the malignant group, no statistically significant difference of CTC counts was found between patients with tumor recurrence and those without (median [IQR], 3.0 [2.0 to 7.5] vs. 3.0 [1.0 to 5.0]; $p=0.102$) (Fig. 3H).

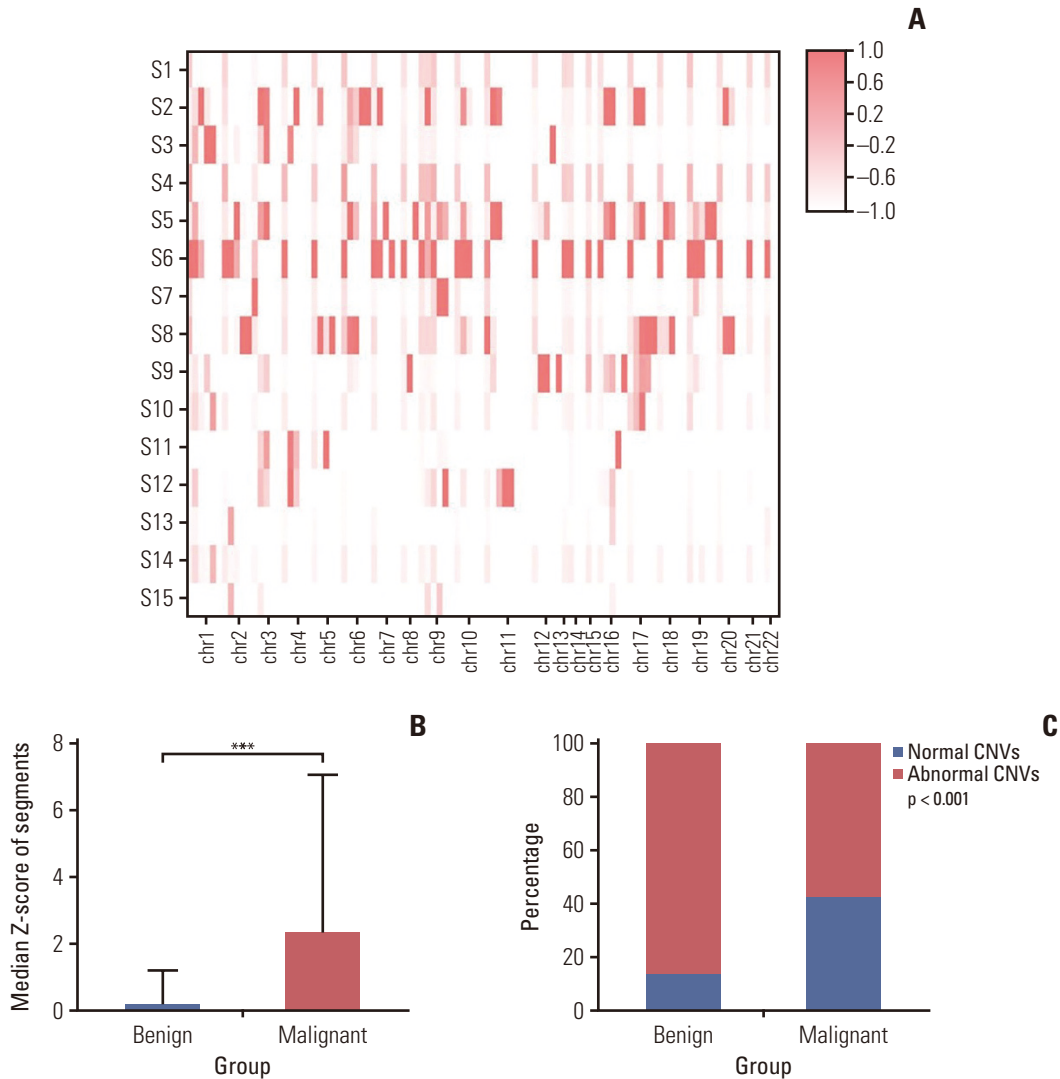


Fig. 4. Low-coverage whole-genome sequencing (LC-WGS)-based copy number variation (CNV) analysis of circulating tumor cells (CTCs) from 15 cases. (A) Z-scores of segments on 22 chromosomes in S1-S15. (B) The comparison of median Z-score of all segments between the malignant group and the benign group. (C) The comparison of abnormal CNVs between the malignant group and the benign group. $|Z\text{-score}| \geq 3$ was regarded as abnormal CNV. Malignant group, CTCs from malignant cases (S1-S8); benign group, CTCs from benign cases (S9-S15). *** $p < 0.001$. p-values were derived from the chi-square test or the Mann-Whitney U test. The median-IQR plot and percent bar chart were used for data visualization. The data were pre-processed by Min-Max Scaling for Fig. 4A. chr, chromosome; IQR, interquartile range.

4. LC-WGS-based CNV analysis

In the cases with gene sequencing, the median values (IQR) of CTC count in the benign and malignant groups were 4.0 (2.0 to 4.0) and 6.0 (2.0 to 7.0), respectively (data not shown). The Z-scores of segments on 22 chromosomes in all cases are shown in Fig. 4A. Overall, CTCs from malignant cases (S1-S8) had more segments with higher Z-scores than those from benign cases (S9-S15). The actual CTC counts and Z-scores of each case are shown in S5 and S6 Tables. The median Z-score of all segments in the malignant group was significantly

higher than that in the benign group (median [IQR], 2.3 [1.1 to 7.1] vs. 0.2 [-0.3 to 1.2]; $p < 0.001$) (Fig. 4B). Considering $|Z\text{-score}| \geq 3$ as abnormal CNV [12,13], the malignant group had significantly more abnormal CNVs than the benign group (42.9% vs. 13.8%, $p < 0.001$) (Fig. 4C).

5. Targeted gene mutation analysis

Fig. 5A shows the landscape of lung cancer-related gene mutations in exon region. Overall, CTCs from most malignant cases had more mutations than those from most benign

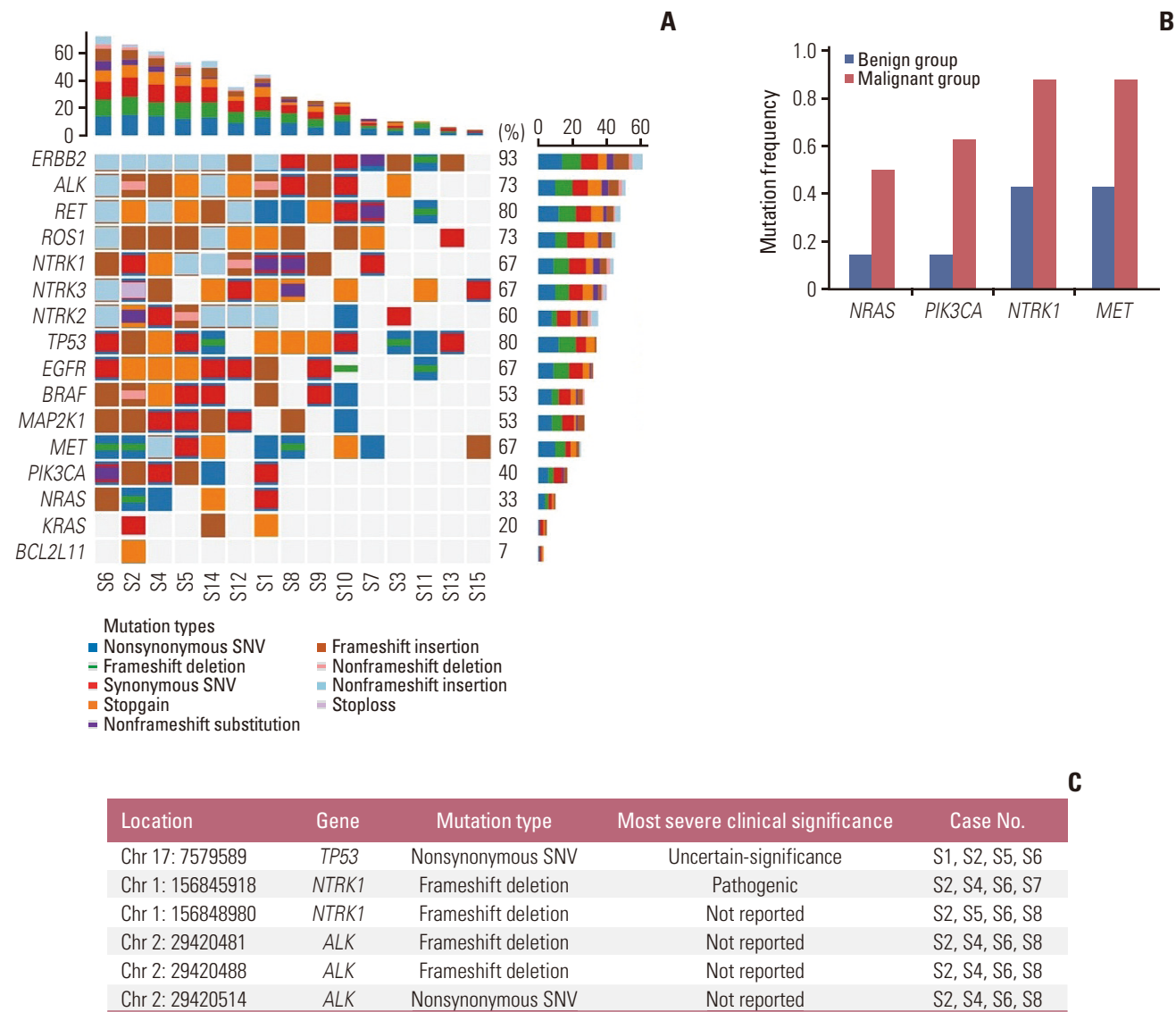


Fig. 5. Lung cancer-related targeted gene mutation analysis of circulating tumor cells (CTCs) from 15 cases. (A) The landscape of mutations in exon region. (B) The difference in exon mutation frequency between the malignant group and the benign group. (C) Shared exon mutations in four malignant samples and the clinical significance in the ClinVar database. Malignant group, CTCs from malignant cases (S1-S8); benign group, CTCs from benign cases (S9-S15). SNV, single nucleotide variation; chr, chromosome.

cases. Mutations in *ERBB2*, *RET*, and *TP53* were found in no less than 80% of all the cases. The most common type of exon mutations was non-synonymous SNV. The frequencies of exon mutations in *NRAS*, *PIK3CA*, *NTRK1*, and *MET* showed relatively large differences between the malignant and benign groups (Fig. 5B). Unfortunately, this study did not find shared exon mutations in the eight malignant cases. The shared exon mutations in four malignant cases were located on *TP53*, *NTRK1*, and *ALK*, including non-synonymous SNV and frameshift deletion (Fig. 5C). Based on the ClinVar database, only the mutation at chr1: 156845918 (*NTRK1*) was

pathogenic, and the clinical significance of SNV located at chr17: 7579589 (*TP53*) was uncertain. The clinical significance of the other mutations was not reported in the database. Furthermore, S7 Fig. shows the comparison between the malignant group and the benign group regarding the number of cancer-risk-related pathogenic or likely pathogenic exon mutations defined in the ClinVar database. No statistically significant difference was observed.

Discussion

In this study, we further confirmed the value of CTCs in the differential diagnosis of benign and malignant PNs, and we found that CTC count might be associated with disease invasiveness. In addition, we showed the genetic signature of CTCs from patients with PNs, which could assist in differential diagnosis. To the best of our knowledge, this is the first study to explore the use of CTC detection by ISET combined with gene sequencing to distinguish between benign and malignant PNs.

It's challenging to find effective diagnostic markers for early cancer because they usually exist in very small amounts [14]. This makes the differential diagnosis of PNs harder, because most of the malignant PNs are in an extremely early stage. Unsurprisingly, we found that 57.2% of the malignant patients had no elevated tumor biomarkers. Regular tumor biomarkers are not adequate for identifying early lung cancer. Therefore, more valuable biomarkers need to be explored. Previous studies have reported the value of different detection methods for CTCs in the evaluation of PNs. However, the diagnostic value or recommendations for the application of CTCs made by these studies were not consistent [15-18]. After expanding the study cohort, we found that the optimal cutoff value for identifying malignant PNs was still 2.5 CTCs per 5 mL of peripheral blood [7]. At this diagnostic cutoff, the specificity reached 78.6%, and the PPV was 92.9%. Considering that the study population consisted of patients with PNs that were clinically suspected to be malignant, a higher PPV could be more beneficial to reduce overdiagnosis by reducing false-positive identification. Furthermore, the sensitivity of diagnosis can be increased to 84.7% when combining with the radiological nodule type and malignant imaging features. This suggests that the combination of chest CT and CTC count to identify the nature of PNs can demonstrate good diagnostic efficacy in clinical practice. Through further classification by radiological nodule type, we found that CTC count showed the best diagnostic efficacy in the SN subgroup, but the diagnostic values in GGN and PSN subgroups were limited. Of note, the small number of benign cases in these two subgroups (three cases in the GGN subgroup, and nine cases in the PSN subgroup) may result in insufficient statistical power. It was difficult to determine whether the diagnostic power of CTC count was accurately reflected. A larger sample size is needed for verification.

We further analyzed the relationship between CTC count and various demographic-clinical characteristics. One study showed that CTC count was associated with the tumor burden of patients with lung adenocarcinoma [19]. We also observed that the patients with relatively later stage disease ($\geq$ stage Ia2) had significantly greater CTC counts. Although

CTCs were traditionally considered to be a late event in tumor progression, recent views have suggested that tumor cells could infiltrate the blood at the beginning in patients with aggressive cancers [6]. Notably, we found that patients with STAS, which suggested greater tumor aggressiveness, had significantly greater CTC counts. Moreover, Zhou et al. [18] reported that CTC count in combination with nodule diameter could effectively differentiate non-invasive cancer (AIS) from invasive cancer (IAC). Although we did not observe statistically significant differences in CTC counts among IAC, MIA, and AIS subgroups, patients with the most aggressive disease (IAC) exhibited relatively greater CTC counts. This suggests that CTC count might have the potential in identifying aggressive tumors in the early stage to guide timely intervention. The possible ability of CTC count to reflect disease invasiveness led us to consider whether it could be used to predict prognosis. However, we did not find a statistically significant difference in preoperative CTC counts between the patients with recurrence and those without. The results of previous studies on the relationship between CTC count before treatment and disease recurrence were inconsistent [20,21]. Further validation with more research data is still required.

It is a challenge to determine the nature of the abnormal cells found by ISET. In this study, with 2.5 CTCs per 5 mL of peripheral blood as the diagnostic cutoff, the low NPV (25.0%) severely limited the diagnostic efficacy. We found that 32.6% of malignant cases had 1-2 CTCs. On the other hand, CTCs were detected in 78.6% of benign cases. The CTCs detected in the benign cases might be false positives caused by abnormal cells with large volume in the blood, such as lymphocytes or monocytes [22]. In addition, these might be genuine tumor cells, which may suggest a potential for progression to malignant disease in the future [23]. In these cases, identifying the origin of CTCs in addition to the CTC count could significantly improve the diagnostic efficacy. CNV is prevalent in various malignant tumors, and is a valuable marker for cancer diagnosis [24]. Recent studies have reported the value of LC-WGS-based CNV analysis in the early diagnosis of various malignant tumors [11,12,25]. The biospecimens used for diagnosis included uterine cavity exfoliated cells, urine exfoliated cells, and circulating free DNA, and CNV analysis was used to identify their origins [11,12,25]. In addition, Ni et al. [26] observed widely-present CNVs in the CTCs from lung cancer patients. These previous studies have suggested that it is feasible to use CNV analysis for identifying the nature of CTCs detected in patients with PNs. Indeed, we found that CTCs from malignant cases had more CNV events than CTCs from benign cases, and abnormal CNVs were also significantly more numerous in the malignant cases. Of note, to avoid selection bias, we selected

cases with various CTC counts for sequencing. The results showed that, in cases with CTC count of 2, CTCs from the malignant cases had more abnormal CNVs than those from the benign cases (S5 Table). Moreover, we found that the percentages of abnormal CNVs were relatively low in most benign cases, even if the CTC counts were greater than 2.5 (S5 Table). This suggests that LC-WGS-based CNV analysis might be an optional method to further evaluate the possibility of the malignant origin of CTCs. The next step should be focus on how to quantify the use of CNV analysis in a larger cohort, and clarify the diagnostic efficacy of combined CTC count with CNV analysis in the differential diagnosis of PNs.

A recent study has revealed that key driver mutations occur in the very early stage lung adenocarcinoma [27]. This suggests the feasibility of using targeted gene sequencing to identify early-stage malignant diseases in patients with PNs. However, the specific susceptibility loci associated with malignant PNs are not well-studied. The reported susceptibility loci in the existing studies are inconsistent [15,28], and the practical significance of these findings remains uncertain. Unfortunately, in this study, no difference between the benign and malignant cases was observed in the three TP53 SNVs (g.7578115 T>C, g.7578645 C>T, g.7579472 G>C) that had been found in our previous study [7]. And we failed to identify any specific susceptibility loci shared by most malignant cases yet absent in benign cases. This makes sense to some extent. Namely, it is speculated that identifying a unique and highly specific pathogenic locus is challenging because lung cancer is a polygenic disease. Drawing on the findings of this study, we considered that SNV/InDel detection of CTCs might not be suitable for differentiating between benign and malignant PNs. The more appropriate research direction for gene mutation detection of CTCs might be guiding precision treatment and dynamic monitoring [29].

This study had some limitations. First, the number of the included benign PNs was relatively small. This also limits the ability to further explore diagnostic value in different subgroups. However, for overall patients, after expanding the study cohort, the diagnostic sensitivity and specificity were similar to those of our former study. This suggested good stability of the detection. Second, sequencing analysis of CTCs still poses technical challenges, and the high cost also limits the applications. However, LC-WGS-based CNV analysis and targeted gene sequencing were selected in this study, which may improve the efficiency and reduce the cost. Third, although we performed sequencing analysis on more cases, it was insufficient to establish the diagnostic model based on CNV analysis. The sequencing analysis in this study was still exploratory; however, the results support the feasibility of LC-WGS-based CNV analysis in CTCs to distinguish between benign and malignant PNs. Finally, by

the end of the follow-up, the number of recurrence cases was small, which was insufficient to draw a conclusion regarding the association of CTC count and postoperative recurrence.

In conclusion, this study evaluated the application value and potential of CTCs in the early diagnosis of lung cancer. To be specific, for patients with PNs, CTC count greater than 2.5 per 5 mL of peripheral blood suggests a higher possibility of malignant disease. The diagnostic accuracy can be improved if combined with CT imaging characteristics. Further LC-WGS-based CNV analysis may be performed to evaluate the possibility of the malignant origin of CTCs, thereby assisting in differential diagnosis. Moreover, CTC count may be associated with the disease invasiveness and potentially help identify aggressive tumors in the early stage.

Electronic Supplementary Material

Supplementary materials are available at Cancer Research and Treatment website (<https://www.e-crt.org>).

Ethical Statement

This study was performed in compliance with the principles of the Declaration of Helsinki, and was approved by the Ethics Committee of the China-Japan Friendship Hospital (2018-170-K124). All of the participants provided written informed consent.

Author Contributions

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Conflicts of Interest

Conflict of interest relevant to this article was not reported.

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